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Abstract

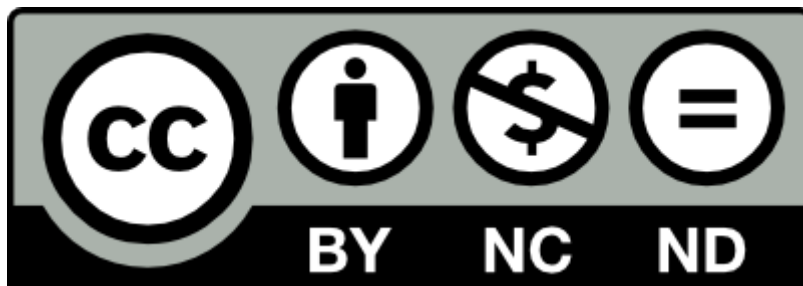
Aim: To develop and evaluate a method to detect circulation in the presence of organized rhythms (ORs) during resuscitation using signals acquired by defibrillation pads.

Methods: Segments containing electrocardiogram (ECG) and thoracic impedance (TI) signals free of artifacts were used. The ECG corresponded to ORs classified as pulseless electrical activity (PEA) or pulse-generating rhythm (PR). A first dataset containing 1091 segments was split into training and test sets to develop and validate the circulation detector. The method processed ECG and TI to obtain the impedance circulation component (ICC). Morphological features were extracted from ECG and ICC, and combined into a classifier to discriminate between PEA and PR. The performance of the method was evaluated in terms of sensitivity (PR) and specificity (PEA). A second dataset (86 segments from different patients) was used to assess two application of the method: confirmation of arrest by recognizing absence of circulation during ORs and detection of return of spontaneous circulation (ROSC) during resuscitation. In both cases, time to confirmation of arrest/ROSC was determined.

Results: The method showed a sensitivity/specificity of 92.1%/90.3% and 92.2%/91.9% for training and test sets respectively. The method confirmed cardiac arrest with a specificity of 93.3% with a median delay of 0s after the first OR annotation. ROSC was detected with a sensitivity of 94.4% with a median delay of 57s from ROSC onset.

Conclusion: The method showed good performance, and can be reliably used to distinguish perfusing from non-perfusing ORs.

Keywords: Circulation detection; Defibrillator; Pulse; Pulse-generating rhythm; Pulseless electrical activity; Thoracic impedance.



Circulation detection using the electrocardiogram and the thoracic impedance acquired by defibrillation pads

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Circulation detection, Pulse, Thoracic Impedance, Pulseless Electrical Activity, Pulse-generating Rhythm, Defibrillator

1. INTRODUCTION

During resuscitation, recognition of cardiac arrest and detection of return of spontaneous circulation (ROSC) are challenging for both lay rescuers and healthcare personnel.^{1,2} The carotid pulse check was the protocol accepted to detect absence or presence of circulation until 1998.³ It was later proven both time-consuming and inaccurate,⁴⁻⁸ and was thereafter, no longer included in the guidelines. Current resuscitation guidelines recommend checking for signs of life during cardiopulmonary resuscitation (CPR), such as purposeful movement, breathing or coughing.^{9,10} They also recommend using the capnogram, generally available only with intubation, as a decision support tool for ROSC detection. An abrupt increase in CO₂ level to normal values (35 to 40 mmHg) has been accepted as an indicator of ROSC.¹¹⁻¹⁵ However, there is still a need for a real-time hemodynamic monitor in a resuscitation scenario.¹⁶

In the context of automated external defibrillators (AEDs), the capnogram is rarely available and the only signals recorded are often the electrocardiogram (ECG) and the thoracic impedance (TI) acquired by defibrillation pads. Both have been shown to include valuable information to discriminate pulse-generating from pulseless rhythms in the presence of an organized rhythm (OR).¹⁷⁻²¹ The TI signal, in absence of artifacts due to chest compressions or ventilations, shows a very small fluctuation (usually less than 0.1 mΩ) with every effective heartbeat. The studies by Losert et al.¹⁷ and Risdal et al.¹⁸ concluded that a combination of features extracted from ECG and TI allows a proper discrimination between pulse-generating rhythms (PR) and pulseless electrical activity (PEA). Cromie et al. used a single feature computed from the spectrum of the processed TI as a clinical marker of circulatory collapse in animals¹⁹ and humans.²⁰ In our previous work, we presented a reliable method to extract the impedance circulation component (ICC) using an adaptive processing scheme, and concluded that morphological features extracted from ICC showed great potential to discriminate between PR and PEA.²¹

In this study, we propose a method that uses the ECG and TI signals acquired by defibrillation pads to detect presence of circulation. The method is intended to be launched when an OR is detected. This could be done (1) automatically, by integrating the method into the shock advisory algorithm (SAA) of an AED or (2) manually, integrating the method into a monitor/defibrillator as a decision support tool which the healthcare professional could activate (e.g. by pressing a softkey). The circulation detector was designed to discriminate PR from PEA segments using a training set

31 and validated with a test set. Finally, a case study was carried out using different complete episodes
32 to assess two applications of the method: confirmation of cardiac arrest by recognizing absence of
33 circulation during ORs and detection of circulation during resuscitation.

2. MATERIALS AND METHODS

2.1. Data materials

The dataset used in this study was a subset of a large out-of-hospital cardiac arrest (OHCA) registry containing 385 episodes maintained by the Tualatin Valley Fire & Rescue (Tigard, Oregon, USA). The episodes, one per patient, were collected using the Philips HeartStart MRx monitor/defibrillator between January 2010 and December 2014. The episodes were visually inspected looking for ECG segments that met the following inclusion criteria: (1) OR with a duration of at least 5 s, and (2) no artifacts due to chest compressions or ventilations. Those OR segments were classified as PEA or PR by three expert reviewers (a biomedical engineer, EAL; an emergency medicine physician, MD; and a cardiologist, LGT) using a majority criterion. The annotation was made in the basis of the visual inspection of the ECG, available clinical information and capnography signal. The clinical information derived from the prehospital record consisted of (1) the time of the first detected ROSC in field defined as a palpable pulse in any vessel for any length of time, (2) information about ROSC loss prior to arrival at emergency department, and (3) outcome (death in the field, expired in emergency department, expired post admission or discharged alive).

Each segment contained the ECG (resolution $1.03\text{ }\mu\text{V}$ per least significant bit with bandwidth 0–50 Hz) and the TI (resolution $0.74\text{ m}\Omega$ per least significant bit with bandwidth 0–80 Hz) signals. The TI was acquired by applying a sinusoidal excitation current (32 kHz, 3 mA peak to peak) between the defibrillation pads.

A first dataset was composed of 1091 segments that met the inclusion criteria and were extracted from 158 episodes recorded until May 2014. The dataset was randomly split into training (60%) and test (40%) sets to develop and validate the circulation detector. The PR and PEA segments extracted from the same episode were all assigned to either training or test sets.

A second dataset, the case study dataset, was used to test the clinical applicability of the circulation detector. For this purpose 33 complete episodes gathered between June and December 2014 were used. In 15 episodes the patient never recovered ROSC and finally expired. The first three artefact-free OR segments from these episodes were used to assess the capacity of the circulation detector to confirm cardiac arrest by recognizing absence of circulation. In 18 episodes patients recovered ROSC and a clinical annotation of ROSC time was available. After ROSC, the

first three segments from these episodes were used to evaluate the ability of the method to detect ROSC during resuscitation.

2.2. Circulation detector

Fig. 1 shows the overview of the detector. First, the ECG signal, $ecg[n]$, and the TI signal, $z[n]$, where n is the time sample number, were digitally processed to (1) extract the ICC, and (2) to compute waveform features that characterize PR and PEA. Then, a classifier was designed based on the extracted features.

2.2.1. Signal processing

The $ecg[n]$ signal was band-pass filtered between 0.5–30 Hz to remove baseline wander and high frequency noise. The instants of QRS complexes, n_{QRS} , were automatically detected using the offline Matlab version of the QRS detector running in the Bexen cardio Reanibex 800 monitor/defibrillator.

The $z[n]$ signal was band-pass filtered between 0.65–7 Hz to suppress high frequency noise and low frequency components below the fundamental frequency of ICC. The resulting signal, $z_p[n]$, was processed using an adaptive scheme based on a recursive least square (RLS) algorithm to suppress the remaining noisy components, obtaining the ICC signal, $\hat{icc}[n]$.^{22,23} A detailed description of the adaptive filtering scheme is provided in appendix A.

2.2.2. Feature extraction and classification

Several waveform features extracted from $ecg[n]$, $\hat{icc}[n]$, and its first derivative, $d\hat{icc}[n]$, were used to build a multivariate logistic regression classifier. The training set was used to develop the optimal classifier, as detailed in appendices B and C. The optimal classifier combined the following six features: mean area of $d\hat{icc}[n]$, mean RR interval, standard deviation of the amplitude of $\hat{icc}[n]$, mean amplitude of the QRS complexes, mean area of $\hat{icc}[n]$, and standard deviation of the amplitude of the QRS complexes.

2.3. Evaluation and statistical analysis

The performance of the circulation detector was assessed using the test set in terms of sensitivity (capacity to correctly detect PR), specificity (capacity to correctly detect PEA), and area under the curve (AUC). Results are presented as median (inter-quartile range, IQR) because their

distributions did not pass the Anderson-Darling normality test. Consecutive diagnoses by the pulse detector within the same patient are correlated, so sensitivity and specificity and their 90% one-sided lower confidence limit were adjusted for clustering within patients using generalized estimating equations within a logistic regression model.²⁴ The calculations were done using the Geepack library²⁵ in the R statistical software.

2.4. Case study

Two applications of the circulation detector were simulated. First, the ability of the detector to confirm cardiac arrest by recognizing the absence of circulation. This simulates a scenario where the rescuer might have the uncertainty of being in front of a cardiac arrest. In such scenario, a decision support that would confirm cardiac arrest would be the most valuable. The specificity of the circulation detector was assessed independently for the three segments, and time to confirmation of arrest was determined. Second, the capability of the detector to detect ROSC during resuscitation. The sensitivity of the circulation detector was evaluated independently for the three segments, and time to the first detection of ROSC was determined.

106 3. RESULTS

107 Table 1 summarizes the number and median (IQR) duration of PEA and PR segments included
108 in the datasets. Overall, 1177 segments from 191 patients with a duration of 8.3 (6.6 – 11.7) s were
109 analyzed. Specifically, 796 PR segments from 99 patients and duration of 7.9 (6.4 – 11.0) s and 381
110 PEA segments from 92 patients and duration of 9.6 (7.0 – 13.6) s.

111 Panels a and b in Fig. 2 shows the $ecg[n]$ and $\hat{icc}[n]$ signals for a PEA and PR segment
112 respectively. Both ECG signals are regular with similar mean RR interval, but the magnitude of
113 the $\hat{icc}[n]$ is greater for PR than for PEA.

114 The circulation detector showed a sensitivity/specificity of 92.1% (90% one-sided lower
115 confidence limit, 88.5%)/90.3% (86.2%) and AUC value of 0.95 for the training set. While, a
116 sensitivity/specificity of 92.2% (87.7%)/91.9% (87.2%) and an AUC value of 0.97 for the test set.

117 In the case study, the specificities for the confirmation of cardiac arrest were 93.3%, 100%, and
118 100% respectively for the three PEA segments of the episodes. In these cases, the time instant
119 when the circulation detector was launched was 0 (0 – 65) s, 122 (102 – 180) s, and 239 (195 – 416) s
120 with respect to the first annotation of an OR. For the detection of ROSC during resuscitation, the
121 sensitivities were 94.4%, 100%, and 100% respectively for the three PR segments of the episodes
122 in the case study. The time instant when the circulation detector was launched was 57 (11 – 38) s,
123 138 (46 – 327) s, and 268 (136 – 393) s with respect to the first clinical annotation of ROSC.

124 Fig. 3 shows the $ecg[n]$ (top) and $\hat{icc}[n]$ (bottom) signals for the first (panel a), second (panel
125 b) and third (panel c) PEA segments of the patient where the first PEA segment was misclassified.
126 Similarly, Fig. 4 depicts the $ecg[n]$ (top) and $\hat{icc}[n]$ (bottom) signals for the first (panel a),
127 second (panel b), and third (panel c) PR segments of the patient where the first PR segment
128 was misclassified.

4. DISCUSSION

In this paper, a method to detect the presence of circulation during resuscitation scenario has been presented. For every OR, the ECG and TI signals acquired by defibrillation pads were analyzed, and the rhythm classified as PEA or PR. Through a case study two potential applications of the method were tested: recognition of cardiac arrest and identification of ROSC.

4.1. The dataset and the gold standard

The circulation detector was developed, tested, and its applications assessed using a large dataset with 1177 PR/PEA segments extracted from OHCA episodes. The gold standard to annotate an OR as PR or PEA played a crucial role. It is accepted that additional information is needed to make that decision. In 2007 Losert et al.¹⁷ used the invasive measurement of the systolic blood pressure as gold standard in an in-hospital setting. Risdal et al.¹⁸ used as gold standard the manual rhythm annotations made by expert reviewers in an OHCA scenario. These annotations were made by visual inspection of the ECG and TI signals looking for fluctuations in the TI correlated with the heartbeats. In our study, we used a comprehensive set of variables to annotate ORs as PEA or PR. Variables included ROSC onset time and ROSC loss intervals annotated in-situ by the rescuers, patient outcome, and ECG and capnography signals. This is, to our knowledge, the most comprehensive set of variables used to date for PEA/PR annotation in retrospective OHCA data.

4.2. The circulation detector

The circulation detector presented good performance, a sensitivity/specificity of 92.2% (87.7%)/91.9% (87.2%) for the test set, improving previously reported results.^{17,18,20} The adaptive scheme used to extract the ICC is determinant as the features extracted from ICC showed great discrimination power (AUC=0.94 for the best single feature, the mean area of the $\hat{d}_{icc}[n]$), and the robustness of the detector increased when ICC and ECG features were combined (AUC=0.97).

4.3. Case study

The results obtained from the case study confirmed the expected good performance of the circulation detector and gave a time-perspective of its application as a support tool to confirm cardiac arrest and to detect ROSC during resuscitation.

When the ability to confirm cardiac arrest was evaluated, only the initial PEA segment of one patient was misclassified, as shown in panel a in Fig. 3. The magnitude of the ICC was significant which led the detector to make a classification error. Nevertheless, the following PEA segments (panels b and c in Fig. 3) were correctly detected. For the rhythm reassessments (2nd and 3rd segments), no PEA segment was misclassified in any episode. The reassessments were carried out at 122 s and 239 s respectively from the instant the OR was detected which reproduces the 2-min CPR cycle followed by a rhythm reassessment of the BLS protocol when an AED is available.

When the capability to detect ROSC during resuscitation was evaluated, only the initial PR segment of one patient was misclassified as shown in panel a in Fig. 4. The magnitude of the ICC was small, in contrast to the following PR segments (panels b and c in Fig. 4) which were correctly detected. No CPR was provided and half a minute passed between the first and second PR segments. Therefore, it could be that the mechanical activity of the heart was not sufficient to cause significant fluctuations in the TI during the first seconds of ROSC. For the rhythm reassessments (2nd and 3rd segments), no PR segment was misclassified in any episode. The reassessments were carried out at 138 s and 268 s respectively from the clinically annotated ROSC time. The presence of hyperventilation observed after ROSC delayed the launch of the detector.

4.4. Action protocol

An action protocol could be defined to integrate the circulation detector into BLS and ALS. In BLS, the detector could be integrated together with the SAA of an AED without varying the current protocol. The detector would be launched in a rhythm analysis interval, where no artifacts due to chest compressions or ventilations are present, and immediately after the detection of an OR by the SAA. In the current ALS protocol there are few spots without artifacts where the detector could be launched because normally several healthcare professionals are involved, and the rhythm assessment is carried out by visual inspection of the ECG while ventilating the patient. Therefore, we propose the detector to be integrated into a monitor/defibrillator as a decision support tool that could be manually activated by the healthcare professional when an OR is observed and presence of a pulse suspected.

4.5. Limitations

The results obtained are closely linked to specific features of signals acquired by the Philips MRx monitor/defibrillator. TI signals acquired by other equipment must first be validated for the

187 purpose of detecting circulation.

188 **5. Conclusions**

189 In this study, a method to detect the presence of circulation has been proposed using the
190 ECG and TI signals acquired via defibrillation pads. The method showed good performance when
191 validated with the test set, and the case study revealed its applicability as a tool to confirm cardiac
192 arrest and to detect ROSC during resuscitation. However, the method must be validated using
193 signals acquired by AEDs, and it should be incorporated and assessed in a real time framework
194 before integrating it into defibrillators.

195 **Ethical Approval**

196 The CPR process files used in this study were collected as part of an effort to develop an
197 airway check algorithm using the capnography signal. Since these raw data files have no identifying
198 information, the Institutional Review Board at the Oregon Health & Science University determined
199 that the proposed activity is not human subject research because the proposed activity does not
200 meet the definition of human subject per 45 CFR 46.102(f).

201 **Conflict of interest**

202 Mohamud Daya is an unpaid consultant for Philips Healthcare.

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210 Appendix

211 A. Adaptive extraction of the ICC

212 The $z_p[n]$ signal was adaptively filtered to extract the $\hat{icc}[n]$ signal using the scheme illustrated
 213 in Fig. 5. The $\hat{icc}[n]$ was modelled using 3 harmonics of slowly changing amplitude and phase^{21,26}
 214 as represented in the following equation:

$$\hat{icc}[n] = \sum_{k=1}^3 A_k[n] \cos(2\pi k n f[n] + \varphi_k[n]) = \sum_{k=1}^3 \hat{a}_k[n] \cos(2\pi k n f[n]) + \hat{b}_k[n] \sin(2\pi k n f[n]). \quad (1)$$

215 where $f[n]$ denotes the instantaneous frequency of the QRS complexes computed from the
 216 instants n_{QRS} .

217 Fig. 5 shows how the RLS algorithm recursively computes the amplitudes of the in-phase and
 218 quadrature components, $\hat{a}_k[n]$ and $\hat{b}_k[n]$, for each one of the three harmonics ($k = 1, 2$, and 3) to
 219 estimate the $\hat{icc}[n]$. The adaptive process of the RLS algorithm is defined by the forgetting factor,
 220 λ , which is basically a weighting factor to ensure that data in the close past are positively weighted
 221 over data in distant past so that the statistical variations of the data are tracked.²³ The training
 222 set was used to optimize the value of λ (which is always close to, but less than one) resulting in an
 223 optimal value of 0.9997. The RLS algorithm has a shorter transient interval (quicker adaptation)
 224 than the LMS algorithm proposed in our previous work.²¹

225 B. Waveform feature selection

226 Up to 16 waveform features, $v_1 - v_{16}$,^{17,18,21} were considered in the optimization process to build
 227 the classifier. The training set was used to find the best feature combination that maximized the
 228 balanced error rate, i.e. the average value of the sensitivity and specificity. This ensures balanced
 229 detection results and avoids biases due to class imbalance in the training set. Features were selected
 230 using a greedy forward subset feature selection approach. The next 6 features were chosen in the
 231 following order:

- **Mean area of $\hat{dicc}[n]$, v_1 :** Area per sample under the absolute value of $\hat{dicc}[n]$.

$$v_1 (\text{m}\Omega \cdot \text{s}^{-1}) = \frac{1}{N} \sum_{n=1}^N |\hat{dicc}[n]| \quad (2)$$

232 where N represents the number of samples of the analyzed segment.

- **Mean RR interval, v_2 :** Mean value of the RR intervals of a segment:

$$v_2(s) = \frac{1}{N_{QRS} - 1} \sum_{i=1}^{N_{QRS}-1} t_{RR_i} \quad (3)$$

233 where N_{QRS} is the number of QRS complexes of the segment, and t_{RR_i} the time in seconds
 234 of i^{th} RR interval.

- **Standard deviation of the amplitude of $\hat{icc}[n]$, v_3 :** Standard deviation of the peak-to-trough amplitude of the fluctuations caused by each effective heartbeat in the $\hat{icc}[n]$:

$$v_3(m\Omega) = \sqrt{\frac{1}{N_{QRS} - 1} \sum_{i=1}^{N_{QRS}-1} (a_i - \mu)^2} \quad (4)$$

235 where a_i denotes the peak-to-trough amplitude of the fluctuation in the i^{th} RR interval, and
 236 μ the mean peak-to-trough amplitude of the fluctuations in the segment.

- **Mean amplitude of the QRS complexes, v_4 :** The mean value of the peak-to-peak amplitude of the QRS complexes detected in $ecg[n]$:

$$v_4(mV) = \frac{1}{N_{QRS}} \sum_{i=1}^{N_{QRS}} pp_i \quad (5)$$

237 where pp_i represents the peak-to-peak amplitude of i^{th} QRS complex, computed as the
 238 difference between maximum and minimum of the $ecg[n]$ subsegment corresponding to the
 239 i^{th} QRS complex.

- **Mean area of $\hat{icc}[n]$, v_5 :** Area per sample under the absolute value of $\hat{icc}[n]$.

$$v_5(m\Omega) = \frac{1}{N} \sum_{n=1}^N |\hat{icc}[n]| \quad (6)$$

- **Standard deviation of the amplitude of the QRS complexes, v_6 :** The standard deviation of peak-to-peak amplitudes of the QRS complexes.

$$v_6(mV) = \sqrt{\frac{1}{N_{QRS}} \sum_{i=1}^{N_{QRS}} (pp_i - v_4)^2} \quad (7)$$

240 C. Classifier

241 The classifier was based on a multivariate logistic regression model using $v_1 - v_6$ features to
242 assign a probability, $P_{\text{PR}}(z)$, to each segment as:

$$P_{\text{PR}}(z) = \frac{1}{1 + e^{-z}} \quad (8)$$

243 with $z = \theta_0 + \theta_1 \cdot v_1 + \theta_2 \cdot v_2 + \theta_3 \cdot v_3 + \theta_4 \cdot v_4 + \theta_5 \cdot v_5 + \theta_6 \cdot v_6$

244 The training set was used to compute the regression coefficients (θ_i). Segments with an assigned
245 $P_{\text{PR}} > 0.5$ ($z > 0$) were classified as PR, while the opposite ($z \leq 0$) was classified as PEA. The
246 imbalance between classes was compensated weighting each sample according to the proportion of
247 samples in each class.

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Figure Legends

- Figure 1 Overview of the circulation detector consisting of three different stages. First, the signal processing stage where the ECG signal, $ecg[n]$, and the TI signal, $z[n]$, are digitally processed to detect the instants of the QRS complexes, n_{QRS} , and to extract the ICC signal, $icc[n]$, and its first derivative, $\hat{d}icc[n]$, respectively. Second, the feature extraction stage where features characterizing the waveform of $ecg[n]$, $icc[n]$, and $\hat{d}icc[n]$, are extracted. Finally, the classifier distinguishes between PEA and PR based on v_i features.
- Figure 2 Examples of PEA (panel a) and PR segments (panel b) where the $ecg[n]$ and $\hat{icc}[n]$ signals are depicted from top to bottom.
- Figure 3 First (panel a), second (panel b) and third (panel c) PEA segments of the patient of the case study where the first PEA segment was misclassified. The $ecg[n]$ signals are presented at the top, while the $\hat{icc}[n]$ signals are depicted at the bottom.
- Figure 4 First (panel a), second (panel b) and third (panel c) PR segments of the patient of the case study where the first PR segment was misclassified. The $ecg[n]$ signals are presented at the top, while the $\hat{icc}[n]$ signals are depicted at the bottom.
- Figure 5 Diagram of the adaptive scheme applying a RLS algorithm used to extract the $\hat{icc}[n]$ from the filtered TI signal, $zp[n]$.

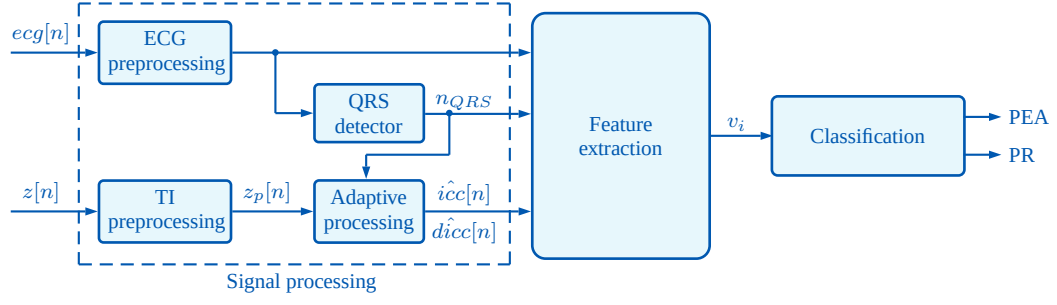


Figure 1: Overview of the circulation detector consisting of three different stages. First, the signal processing stage where the ECG signal, $ecg[n]$, and the TI signal, $z[n]$, are digitally processed to detect the instants of the QRS complexes, n_{QRS} , and to extract the ICC signal, $\hat{icc}[n]$, and its first derivative, $\hat{dicc}[n]$, respectively. Second, the feature extraction stage where features characterizing the waveform of $ecg[n]$, $\hat{icc}[n]$, and $\hat{dicc}[n]$, are extracted. Finally, the classifier distinguishes between PEA and PR based on v_i features.

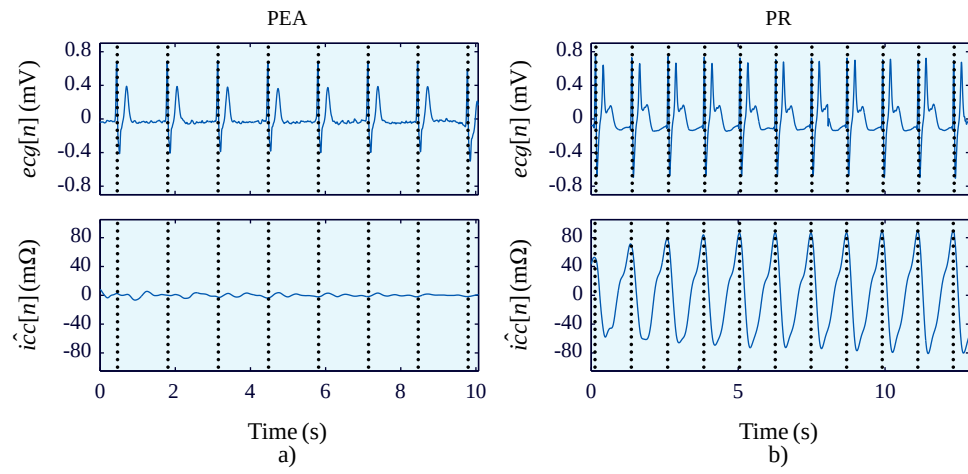


Figure 2: Examples of PEA (panel a) and PR segments (panel b) where the $ecg[n]$ and $\hat{icc}[n]$ signals are depicted from top to bottom.

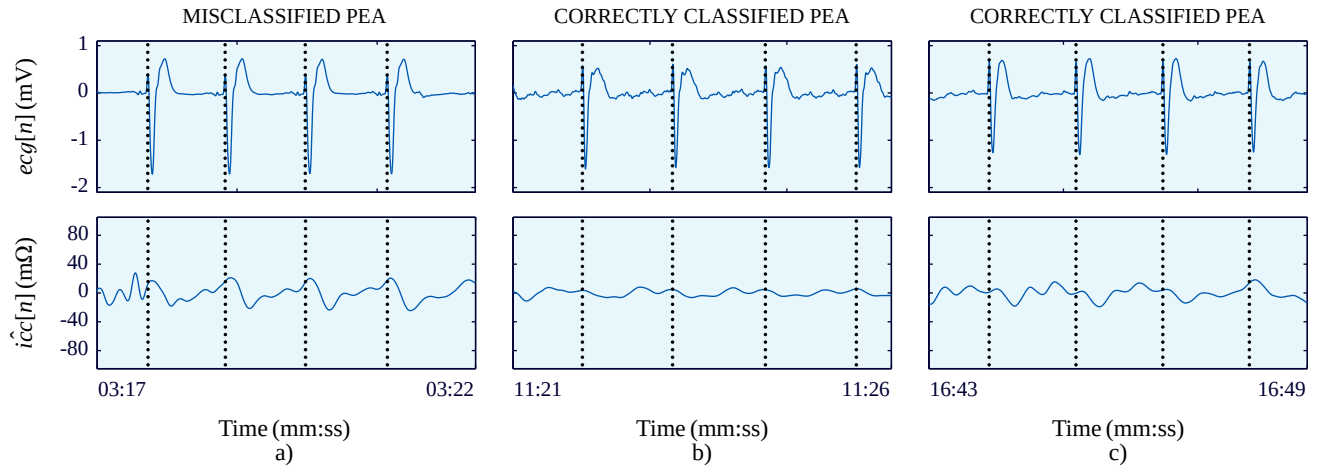


Figure 3: First (panel a), second (panel b) and third (panel c) PEA segments of the patient of the case study where the first PEA segment was misclassified. The $ecg[n]$ signals are presented at the top, while the $icc[n]$ signals are depicted at the bottom.

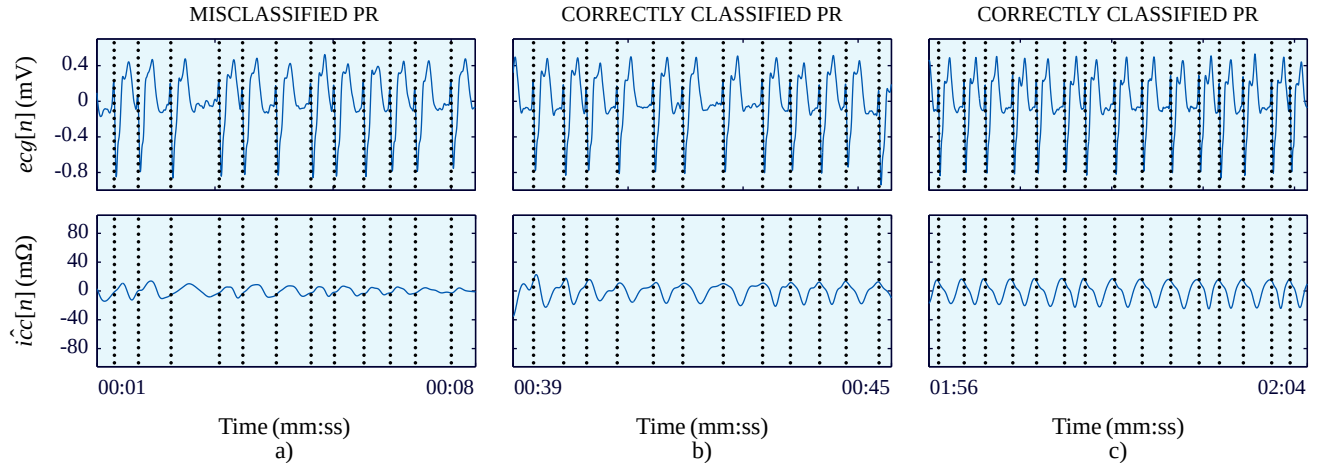


Figure 4: First (panel a), second (panel b) and third (panel c) PR segments of the patient of the case study where the first PR segment was misclassified. The $ecg[n]$ signals are presented at the top, while the $i\hat{c}c[n]$ signals are depicted at the bottom.

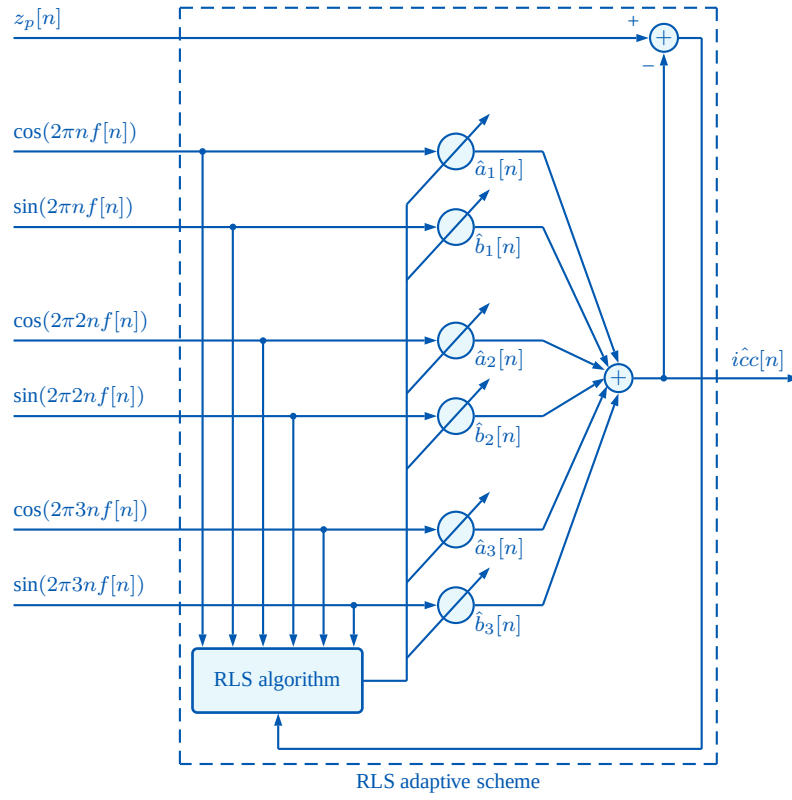


Figure 5: Diagram of the adaptive scheme applying a RLS algorithm used to extract the $\hat{icc}[n]$ from the filtered TI signal, $z_p[n]$.

329 Table Legends

330	Table 1	Summary of segments grouped in training, test and case study datasets.
331		The number of patients is indicated in parenthesis. Duration of the
332		segments is expressed as median (IQR).

Table 1: Summary of segments grouped in training, test and case study datasets. The number of patients is indicated in parenthesis. Duration of the segments is expressed as median (IQR).

Database	PEA		PR	
	No. segments	Duration (s)	No. segments	Duration (s)
Training	205 (81)	9.8 (7.4 – 14.3)	449 (81)	7.8 (6.5 – 10.9)
Test	136 (77)	9.3 (6.7 – 13.8)	301 (77)	8.2 (6.5 – 11.3)
Case study	40 (15)	8.9 (6.1 – 11.5)	46 (18)	7.4 (5.7 – 8.6)